Decomposing the GP Advantage in Offline Model-Based Optimization

Anonymous submission

Abstract

Offline model-based optimization ships a surrogate and an optimizer as a single pipeline and reports a single number, so when GP-LCB beats a neural ensemble, nothing in the record says which component earned the win—a gap the subfield's 2026 survey concedes. We run the design the question requires: three surrogate classes crossed with three search routines under one shared protocol, decomposed by two-way ANOVA. Building the grid surfaced five confounds specific to offline-MBO surrogate comparison (target scaling, an unequal candidate/oracle rule, and an $11.8\times$ query-budget imbalance), each traced to a scoring-path line, signed, and pre-registered with a kill criterion. Removing the two protocol confounds moves the surrogate effect up, not down (η^2_{surr} from 0.367 uncorrected to 0.405 $[0.290, 0.556]$), against every audit we could find; seven tasks cannot resolve the level, so we claim the direction, not a point value. Three findings anchor the decomposition. First, on the synthetic grid, the surrogate$\times$optimizer interaction—which a one-factor-at-a-time study cannot estimate—excludes zero in all four engine corners and exceeds the optimizer main effect in eleven of twelve corner$\times$normalizer combinations: which optimizer to use depends on the surrogate. Second, the surrogate that loses is the more accurate one—the ensemble's held-out RMSE beats the GP's at every width, while its acquisition function ranks designs it invented above real designs it was handed (on Branin, in all thirty seeds). Third, on Design-Bench the surrogate effect is undetectable at this statistical power—never reaching equivalence—partly because several cells freeze at the dataset reference, a benchmark-construction defect any attribution study must screen for first. Any η^2 headline is a joint artifact of five operating-point coordinates; we supply the taxonomy, the protocol, and the engine stamp that make one reproducible.

1 Introduction

Offline model-based optimization (MBO) seeks a design x maximizing an expensive black-box f from a fixed dataset $\mathcal{D}$, with no further queries to f during search (Trabucco et al. 2022; Kim et al. 2026). The dominant recipe fits a surrogate $\hat{f}$ and optimizes it; because optimization pushes

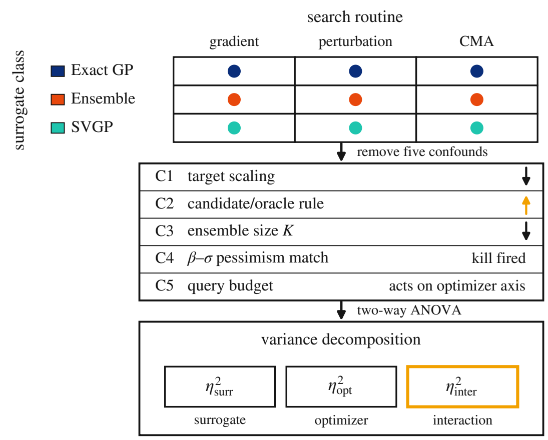

Figure 1: The design: a crossed surrogate$\times$optimizer factorial (three surrogate classes, three search routines, one shared protocol), five confounds specific to offline-MBO surrogate comparison removed, then decomposed by two-way ANOVA into the two main effects and the interaction the factorial exists to estimate.

designs where $\hat{f}$ is unconstrained, the surrogate is made conservative, by a learned regularizer (Trabucco et al. 2021) or a pessimistic acquisition such as the lower confidence bound (LCB) $\mu(x) - \beta\sigma(x)$ (Srinivas et al. 2012). A common intuition holds that Gaussian-process LCB (GP-LCB) beats neural-ensemble LCB at low dimension because the GP posterior is better calibrated. But the two pipelines differ in *two* coupled factors: surrogate class (GP posterior vs. ensemble disagreement) and acquisition optimizer, since GPs are usually optimized by perturbation or quasi-Newton restarts and neural surrogates by gradient ascent on x. Every method in this lineage ships both and reports one number, leaving open which the number credits.

The attribution question is open, by the field's own account. The subfield's 2026 survey states that existing benchmarks "often emphasize overall optimization performance without clarifying whether observed gains stem from superior surrogate modeling, improved optimization strategies, or mere chance" (Kim et al. 2026). That takes a design nobody has run: both axes varied against each other under one shared protocol, the variance decomposed between them. Existing systematic studies fix one axis. The most thorough is *broader* than ours on the surrogate axis precisely because it fixes the other: a seven-class Bayesian neural-network comparison holding Monte-Carlo expected improvement fixed, run *online*, decomposing nothing (Li, Rudner, and Wilson 2024). A neighbouring line varies the *training objective* at fixed architecture, a ranking loss (Tan et al. 2025) or a conservative regularizer (Trabucco et al. 2021): a different factor from model class, since a GP posterior cannot be instantiated without its marginal-likelihood fit. Offline RL's closest re-evaluation likewise varies uncertainty-penalty heuristics at a fixed policy optimizer (Lu et al. 2022). A design that fixes a factor cannot estimate it, so a seven-class comparison at one fixed acquisition rule is not a smaller version of this experiment but a transposed one. We run the crossed surrogate$\times$optimizer factorial (Figure 1) and report the two-way decomposition; no prior offline-MBO work reports one, and the nearest near-miss is close enough to name.[1] Our reference point is PGS, which reports per-task results and claims surrogate class dominates but computes no decomposition (Chemingui et al. 2024); we reproduce its Table 1, compute the decomposition ($\eta^2_{\text{surr}} \approx 0.37$ on the uncorrected engine), then audit it.

The contribution. The instrument, and the confound set that running it exposed. An instrument inherits every confound in the grid it measures, and building the crossed grid surfaced five specific to offline-MBO surrogate comparison, each traced to a line of the scoring path. The contribution is that composition: the confounds, the protocol that removes them, and what the decomposition says once they are gone.

Contributions.

1. **Five named confounds and a removal protocol**, each traced, signed, and pre-registered (Sections 3–4).
2. **A surrogate$\times$optimizer interaction**, the term the design exists to estimate (Section 4).
3. **An operating-point thesis**: five coordinates, or the headline is not reproducible (Section 4).
4. **A mechanism narrowed by elimination**: seven controls, none survives (Section 5).
5. **A Design-Bench null with its power stated**, never equivalence (Section 6) (Agarwal et al. 2021).

Scope. No field reversal: that the statistical model often matters more than the acquisition heuristic is the standard BO review's decade-old organizing doctrine (Shahriari et al. 2016), and our optimizer-axis result falsifies the *local* premise of one recent paper (Chemingui et al. 2024) and nothing wider. We claim no mechanism discovery, no equivalence on Design-Bench, no result about ensemble size.

[1]Table 3 of Tan et al. (2025): a genuine 9×2 crossed grid in offline MBO with a clean 4×2 optimizer-on-trained-model sub-grid, but it crosses training *loss* against method, not model class, and its method axis bundles model family with search paradigm, so it is not commensurable with ours. One further title names both axes (Elsayed and Lacor 2014) but its primary text was unobtainable, and robust-parameter-design methodology is characteristically sequential, which would fail the crossing requirement.

2 Background and Related Work

LCB, pessimism, and coverage. GP-LCB $\mu - \beta\sigma$ has regret guarantees under GP assumptions (Srinivas et al. 2012); deep ensembles (Lakshminarayanan, Pritzel, and Blundell 2017) give a cheap uncertainty proxy with none. The pessimism principle behind conservative offline methods is that a valid lower bound prevents the optimizer exploiting model error (Jin, Yang, and Wang 2021); we make that premise measurable. Conformal methods give distribution-free coverage, including inside the BO loop (Angelopoulos and Bates 2023; Stanton, Maddox, and Wilson 2023); we use coverage of the LCB premise only as a *measurement*, where the optimizer operates, and Section 5 shows it separable from outcome.

Offline MBO and its evaluation. The conservative-surrogate lineage runs from Conservative Objective Models (Trabucco et al. 2021) and its CQL-style regularizer (Kumar et al. 2020) through generative surrogates (Brookes, Park, and Listgarten 2019; Krishnamoorthy, Mashkaria, and Grover 2023), all holding the search rule fixed while varying the model. Optimizer-as-contribution work (Chemingui et al. 2024) proposes a method rather than decomposing, and its motivating premise—that the field improves surrogate models while using fixed search strategies—is the one *local* claim our design falsifies. Design-Bench (Trabucco et al. 2022) standardizes tasks and documents the confounded status quo. Naming unreported choices, giving the protocol that removes them and showing the ranking move is the established reality-check genre, whose shape we claim no credit for (Henderson et al. 2018; Agarwal et al. 2021; Balduzzi et al. 2018; Ferrari Dacrema, Cremonesi, and Jannach 2019; Musgrave, Belongie, and Lim 2020; Lucic et al. 2018); in offline optimization SOO-Bench (Zhu, Qian et al. 2025) stress-tests optimizer *stability*, not attribution. The online-BO controlled study (Li, Rudner, and Wilson 2024) is broader on the surrogate axis but silent on search and attribution; the one-way analogue is fANOVA at a fixed search run (Hutter, Hoos, and Leyton-Brown 2014), and of the two-axis precedents none decomposes, the closest naming fANOVA and *declining* it (Liang et al. 2021; Moosbauer et al. 2022).

3 Experimental Grid and Confounds

The grid. The crossed grid (Figure 1) runs 7 synthetic tasks (Branin-2D through Griewank-30D, drawn once at seed 0) at 30 seeds and 7 Design-Bench tasks at 16 seeds. **Surrogates:** a $K{=}5$ ensemble of two-hidden-layer ReLU MLPs (width 96), with μ, σ the member mean and standard deviation; an *exact GP* (BoTorch `SingleTaskGP`, marginal-likelihood fit on a score-biased subsample); and a *sparse variational GP* (SVGP). **Optimizers:** 100 Adam steps on x; 5 rounds of Gaussian hill-climbing; CMA-ES. All three surrogates are differentiable, so all nine cells are realizable (configuration in the supplement). Every cell maximizes the LCB $\mu(x){-}\beta\sigma(x)$ at $\beta{=}2$, scored by the oracle on its 128 returned designs at the 100th percentile. Per task we min–max normalize the nine cell means and take a two-way ANOVA; η^2_{surr} and η^2_{opt} are the variance fractions of the two main effects, with task-and-seed hierarchical bootstrap intervals ($B{=}10{,}000$).

Two details there are engine state: the exact GP's default covariance is ARD RBF under the BoTorch pinned for the synthetic grid but ARD Matérn-$5/2$ under the older Design-Bench version, and the GPs alone spend per-run marginal-likelihood budget a matched-tuning arm freezes to zero (the surrogate effect retains 75%). Neither is recoverable from a method description that does not stamp its engine. Against that grid we name five confounds, each a specific line of the scoring path rather than a judgment call and each carrying a signed effect (Table 1; four-corner decomposition, Supp. Fig. S1).

Confounds 1–2: target scaling and the candidate/oracle protocol. The ensemble regresses on raw targets spanning -1560 to $+34$ while both GPs z-score theirs (`mbo.py:34,157-162,181-187` vs `mbo.py:293,379-381`); correcting this alone moves η^2_{surr} from 0.367 $[0.254, 0.559]$ to 0.283 $[0.186, 0.444]$. It is a measurement confound, not a bug: the prior analysis states a min–max normalization that exists only in the analysis path, never in training. (A matched-tuning control, freezing per-run marginal-likelihood budget, reaches nearly the same value from a different axis; whether the two are independent our data cannot settle.) Separately, two of three optimizers propose 256 designs and report the oracle-selected best 128 while the third proposes 128 (`mbo.py:35-40,249-255,269-271`), so the reported percentile is two estimands in one column; correcting that alone moves η^2_{surr} to 0.450 $[0.312, 0.649]$. It is the strongest of the five: the protocol the prior analysis declares it holds identical is the identifiability license for the whole comparison, and the license is false.

Confound 3: ensemble size K. K is a confound to control, never a mechanism, and its two halves are read together. *Sensitive*: η^2_{surr} is 0.326, 0.366, 0.408, 0.389 at $K = 2, 3, 5, 10$, peaking at exactly the $K{=}5$ the field fixes by convention (Lakshminarayanan, Pritzel, and Blundell 2017) (Supp. Fig. S1, right). *Non-reversing*: at $K{=}2$ the ensemble marginal is 0.283 against the exact GP's 0.767 and SVGP's 0.739, so our pre-registered kill criterion—"if the ensemble marginal at $K{=}2$ exceeds the GP's, the headline reverses"—did not fire. The prior supplement's 0.95 at $K{=}2$ does not reproduce on the audited engine, which gives 0.283 there. (Confound 1's 0.283 is a variance-explained statistic, this one a cell marginal; neither corroborates the other.) Three of our four K values lie in the $K \in \{2, 5, 10\}$ over which ensemble surrogates were found robust in online BO (Li, Rudner, and Wilson 2024), so we claim no priority on the range; but that work asks whether ensemble *performance* is robust to K (it is), we whether the variance *attributed to the surrogate axis* is (it is not), and only the second bears on a decomposition.

Confounds 4–5: β–σ mismatch and search-intensity budget. We registered that a shared $\beta{=}2$ delivers different effective pessimism across surrogate classes, to be killed "if the ratio is $\sim$1." *That kill fired*: the median $\sigma_{\text{GP}}/\sigma_{\text{ens}}$ is 1.19 at matched $K{=}5$ (1.21 pooled over K), so the fifth unmatched hyperparameter does not exist grid-wide and we withdraw the aggregate claim. What survives is per-task: the ratio spans 0.07 (Branin-2D) to 1.44 (Styblinski-5D, Rastrigin-15D), a 20-fold spread a single multiplier cannot track. That a fixed multiplier on an uncalibrated interval is not comparable across model classes is owned by the interval-validity literature (Dewolf, De Baets, and Waegeman 2022), with a sharper within-class result beside it in offline RL, where sharing pessimistic targets across ensemble members can *invalidate* nominal pessimism outright and flip the bound's sign (Ghasemipour, Gu, and Nachum 2022); that is within-class, so we claim only the offline-MBO acquisition-comparison application, in its narrowed per-task form. Budgets, meanwhile, are unmatched by $11.8\times$ across the optimizer axis, *measured* not derived: gradient $51{,}456$, perturbation $4{,}352$, CMA-ES median $6{,}528$. Three corrections travel with that: the previously estimated "$6\times$–$59\times$" spread was never measured and CMA's nominal 3000 budget is a cap that rarely binds; the proposal-count half is already removed by Confound 2, so the surviving inequality is in queries alone; and the audited trust region is inactive in the main grid, so equalizing queries isolates budget. Matching every optimizer at $Q{=}51{,}456$ (none of 630 cells over 5% off) moves the optimizer main effect from our unmatched η^2_{opt} of 0.005 to 0.038 $[0.003, 0.123]$, an eightfold understatement we disclose; the same matching *raises* the optimizer effect on Design-Bench too (Section 6), so this confound is measured out of both suites.

Table 1: The five confounds. Signed effect is on η^2_{surr} (synthetic, $n{=}7$, 30 seeds) except Confound 5, which acts on η^2_{opt}. The two *protocol* confounds move the headline against the ensemble-size confound; the net is Section 4.

Confound	What it breaks	Signed effect
1. Target scaling	ensemble trains on raw y	$0.367 \to 0.283$
2. Candidate/oracle	two estimands, one column	$0.367 \to 0.450$
3. Ensemble size K	headline fixed at the peak	0.326–0.408
4. β–σ match	per-task, not aggregate	ratio 0.07–1.44
5. Query budget	optimizer = search effort	$0.005 \to 0.038$

4 De-Confounding Protocol and Results

The protocol. Standardize the ensemble's regression targets. Equalize the candidate proposal count across optimizers and impose one selection rule, so the oracle never chooses the reported set. Hold K, β and the query budget fixed and *report them*. Stamp the engine (library versions, flags, commit) with every number. Switching the first two on and off gives four corners (Supp. Table S3).

The correction strengthens the surrogate effect. The named audits in this genre each report a claimed advantage shrinking: an LSTM baseline overtaking newer architectures once hyperparameters are controlled (Melis, Dyer, and Blunsom 2018), the GAN (Lucic et al. 2018), metric-learning (Musgrave, Belongie, and Lim 2020) and neural-recommender (Ferrari Dacrema, Cremonesi, and Jannach 2019) audits. This one strengthens it. Correcting the two protocol confounds moves the surrogate advantage *up*, from our decomposition of PGS's Table 1 ($\eta^2_{\text{surr}} \approx 0.37$) (Chemingui et al. 2024) to 0.405 $[0.290, 0.556]$. The corrections are not additive: each alone moves the headline about 0.08 (down for target scaling, up for candidate/oracle), yet together they land 0.038 above the uncorrected corner, so 0.405 is a measured joint level, not a sum; and the ensemble-size confound moves it the other way, so the strengthening is the net of the two protocol confounds, not a property of de-confounding in general. We found no ML de-confounding audit reporting a corrected variance-explained statistic above its own published value. Two partials: an ImageNet replication reports a relative slope above one $(1.69/1.11)$ but no variance-explained scalar, its headline an 8–11 point drop (Recht et al. 2019), and deep-RL reanalysis moves effects both ways (Agarwal et al. 2021). The direction is not surprising, since a removed confound can suppress as easily as manufacture (Bressan 2019) and at scale corrections run up 19 against down 14 (Maassen et al. 2020); the claim is unprecedented as an instance, not anomalous.

Corner resolution at $n{=}7$ tasks. We pre-registered that it would not resolve, and it did not. The four corner intervals have widths 0.26–0.34 against a corner range of 0.167, sharing a common region $[0.312, 0.443]$ every one contains (Supp. Fig. S1). So 0.405 never appears without its interval, and no corner is asserted to differ from another, though the omnibus rejects in all four ($p \leq 1.7 \times 10^{-3}$). The bootstrap is validated against the reproduction target's precision: our uncorrected corner's width 0.305 matches the 0.32 recorded for it (Chemingui et al. 2024).

The β-dependence is the firm result. η^2_{surr} rises monotonically with pessimism (0.184 $[0.085, 0.354]$, 0.214, 0.282, 0.406 $[0.285, 0.564]$, 0.408 at $\beta = 0, 0.5, 1, 2, 5$), and the $\beta{=}0$ and $\beta{=}2$ intervals barely overlap where the corner intervals do not separate at all (Figure 2). This is the axis the seven tasks come closest to resolving. The query budget goes further: η^2_{surr} more than doubles from 0.243 at $Q{=}4{,}352$ to 0.526 at $Q{=}51{,}456$, those intervals *disjoint* ($[0.189, 0.355]$ vs $[0.421, 0.719]$), the only outright separation here, with a ranking flip. That axis is not ours to claim: budget-dependent benchmarking is a decade-old category (Hansen et al. 2021), and bad models outperforming good ones given enough budget dates to 2018 (Lucic et al. 2018). The *magnitude* of any headline η^2_{surr} is therefore a joint artifact of five operating-point coordinates—K, β, query budget, engine corner, and the per-task normalizer—so a headline stating fewer is not reproducible. We add the normalizer because every η^2 here is on per-task min–max normalized cell means whose range is set by two of the nine arms, an endogenous form known to be outlier-exploitable (Jordan et al. 2024), ranking-flipping at its origin (Bellemare et al. 2013), and pool-dependent as rank statistics are (Balduzzi et al. 2018). This strengthens the decomposition: *within this synthetic grid* the *direction* is invariant to all four, the GP marginals sitting at 0.66–0.85 against the ensemble's 0.17–0.35 across every β, K and corner, never crossing, smallest gap 0.32. Only the size moves, and that invariance does not travel: on Design-Bench the same direction is not detectable at this power (Section 6).

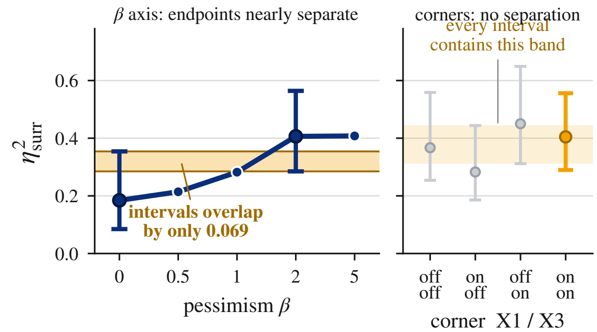

Figure 2: β nearly resolves where the engine corners do not. Left: η^2_{surr} rises monotonically with pessimism, the $\beta{=}0$ and $\beta{=}2$ intervals overlapping by only 0.069 (intervals drawn at those two endpoints only). Right, on the same scale: the four corner intervals overlap completely. Synthetic grid, 7 tasks, 30 seeds, $B{=}10{,}000$.

The interaction is the term the design exists to estimate. η^2_{inter} is 0.165, 0.146, 0.152, 0.160 across corners, intervals $[0.107, 0.260]$, $[0.088, 0.249]$, $[0.094, 0.230]$, $[0.087, 0.236]$: excluding zero in all four, at four to thirty-three times the optimizer main effect (η^2_{opt} 0.013, 0.036, 0.006, 0.005), surviving bias correction (0.134–0.156). It is the most stable quantity in the paper, its range across corners 0.018 against the headline's 0.167.

Two things follow. It is what the crossed design buys, since one-factor-at-a-time experimentation cannot estimate interactions (NIST/SEMATECH 2012; Box, Hunter, and Hunter 1978), which is why the closest two-axis precedent declines the analysis (Moosbauer et al. 2022). And it is why a small η^2_{opt} licenses "optimizer choice explains little *marginal* variance" but never "arbitrary": optimizer choice has a small marginal and a substantial conditional effect, textbook for a factor that is a minor main effect yet dominates the interactions (NIST/SEMATECH 2012). We concede the precedent: interaction reporting is routine in functional ANOVA (Hutter, Hoos, and Leyton-Brown 2014); what we find no instance of is one between *surrogate model class* and *search routine* as independently manipulated pipeline components.

The bound: this is an existence-and-ordering claim under a disclosed normalizer, not the stable quantity "0.15." Across four corners and three normalizers (supplement), $\eta^2_{\text{surr}} > \eta^2_{\text{opt}}$ and η^2_{surr} largest in *all twelve*; the interaction exceeds the optimizer main effect in *eleven*, inverting only in the on/off corner under rank normalization, which carries the largest optimizer effect. Rank roughly halves the interaction throughout. Second, suite-specific: on Design-Bench η^2_{inter} is only 0.007–0.036 and subordinate to η^2_{opt}, which we cannot explain. Frozen cells are a candidate, but Design-Bench's own η^2_{opt} of 0.05–0.17 shows the axes are not all flattened, so the scoping is an unexplained limit, not a resolved one.

5 Isolating the Source of the Gap

Seven controls could each explain the GP–ensemble gap; none does (Table 2). Two were built to confirm accounts of our own and refuted them. Two scopes travel with the count:

Table 2: The seven controls on the GP–ensemble gap; none closes it (audited synthetic grid, β=2, K=5, 30 seeds; detail in the supplement). Elimination 2 supports that the gap *does not close*, not that it is identical at w=1024 (its least precise point), and answers the width objection at practical widths only: the NTK limit is asymptotic, so the depth-driven form remains open.

#	Rules out	Decisive evidence
1	pessimism / σ	61% of the advantage survives at β=0; cross-β gap 0.319 vs 0.525 *(prose)*
2	finite width	gap flat over 10.7$\times$ width, 0.480 $\rightarrow$ 0.476, 99.1% retained (Supp. Fig. S2)
3	predictive accuracy	ensemble RMSE below the GP's at every width; tie-cells 0.375 (registered primary), 7/7 tasks (post-hoc); still loses by $\sim$0.48; NLL is calibration (Supp. Fig. S2)
4	search budget	more budget *widens* the gap; ensemble marginal 0.361 $\rightarrow$ 0.240 while both GPs rise (observation, not preregistered)
5	mean roughness	roughness cut up to 98% both ways, gap widened, no closing interval; GP-roughening arm void
6	premise coverage	own-proposal coverage $\rightarrow$0.998, gap unmoved; ρ(cov, score)=0.29
7	distance / inflation	distance and inflation both null; loss real, -1.406 $[-2.803, -0.375]$ at matched distance *(prose)*

capacity is eliminated at fixed depth, σ at this ensemble construction. Eliminations 1 (σ/pessimism) and 7 (the oracle-value localization) carry the weight and are developed below.

Elimination 1: it is not σ. With pessimism removed (β=0, pure posterior-mean maximization), 61% of the GP's advantage survives and the interval excludes zero: the marginal gap is 0.319 $[0.196, 0.460]$ at β=0 against 0.525 $[0.406, 0.614]$ at β=2, both on one β-invariant normalizer (per-task min–max fit once over pooled cell means of both, refit inside each resample). The increment is itself significant (0.203 $[0.007, 0.396]$, p=0.020): base-plus-amplification, pessimism *amplifying* a mean-quality base rather than creating it, refuting any independence from pessimism. It passes narrowly, 0.319 clearing its pre-registered collapse threshold of 0.263 by a margin inside the interval. Our strongest cross-check: a per-task z-score gives 0.904 against 1.473 standard deviations, a ratio of 0.614 against 0.608 on min–max, so this is a measurement, not a normalization artifact. On our tasks σ also tracks distance more than error, correlating with absolute error at only $\rho \approx 0.07$ but with k-NN distance at $\rho \approx 0.26$, positive on all seven tasks; the prior analysis called σ uninformative by measuring it against the wrong target. That *contrasts* with the distance-aware literature, where deep ensembles are exactly the class characterized as *not* distance aware (Liu et al. 2020; van Amersfoort et al. 2020) (classification-only results). One alternative we do not exclude: our σ is the standard deviation across K squared-error members, reported to underestimate uncertainty (Lakshminarayanan, Pritzel, and Blundell 2017), so this removes σ *as constructed here* and a likelihood-trained variant stays open.

Elimination 7: it is not distance, nor surrogate inflation. Next hypothesis: the ensemble's optima sit further off-support, where its mean invents maxima, and over-predict more. Instrumenting every returned optimum with its 10-NN distance and over-prediction (168 cells, 30 seeds), both are null: $+0.116$ $[-0.004, 0.333]$ further out and $+0.043$ $[-0.652, 1.085]$ more inflated, both covering zero (the distance is one task: Branin-2D $+0.743$, the other six -0.022 to $+0.060$). The loss is not: there the ensemble is worse in *true oracle value* by -1.406 $[-2.803, -0.375]$ in units of $\mathrm{sd}(y_{\mathcal{D}})$, excluding zero, in four of five distance bins. The landscape law is real (across 5,040 optima, distance predicts over-prediction ρ=$+0.758$ and true loss ρ=-0.818) but does not discriminate (Supp. Fig. S3): at matched median distance (0.87/0.86/0.84) the classes differ in outcome anyway. Off-support maxima are inflated and worthless, a property of the problem rather than the ensemble. That law is not ours; Fannjiang and Listgarten (2020) articulated it, oracle-based design querying the oracle where its training data does not reach, so outputs, *including uncertainty*, become unreliable off-support. Our measurement confirms it and shows it *insufficient* here: off-support degradation is shared by every class, so it cannot explain a difference *between* classes. The story is right about where surrogates fail, silent about which maximizers each class admits once there. That degradation has a quantitative form elsewhere, along curves whose coefficients depend on the proxy's class (Gao, Schulman, and Hilton 2023); Elimination 7 moves it onto architecture class, not model scale.

The gap survives seven controls. Budget even corroborates the mean-geometry reading, the ensemble getting relatively worse as both GPs convert budget into score. The seventh adds a positive localization: the loss is real and sits at the returned optimum's oracle quality while every landscape property is matched between the classes. What differs is *which* off-distribution maximizers a surrogate's mean admits, not how rough the means are or how far the optima lie. That is a diagnosis, not a mechanism with a causal test. A skeptic could read this and the interaction as one phenomenon counted twice, but the geometry measurements do not read off the optimizer axis—the σ-distance correlation is a property of the surrogate alone and the matched-distance gap is pooled across optimizers—whereas the interaction is a surrogate$\times$optimizer decomposition, so the mechanism is one phenomenon triangulated by two independent instruments. (A second positive arm, prior-mean reversion, also returned an elimination; supplement.)

Is the advantage conditional on the optimizer? In raw oracle units the exact-GP–ensemble gap is smaller under perturbation than under *both* gradient and CMA on all seven tasks, closing entirely on Ackley-20D, but our one discriminating case points to a floor effect rather than a conditional advantage: on Styblinski-5D, where perturbation is the grid's *best* optimizer (GP with perturbation at 35.9), the attenuation is *weakest* (supplement). The descriptive fact is solid;

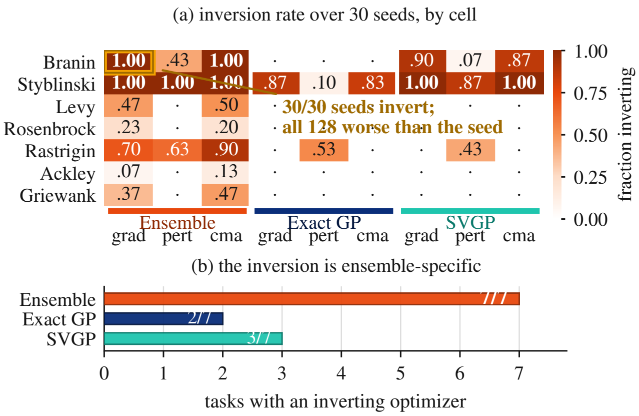

Figure 3: The ensemble's LCB ranks designs it invented above real designs it was handed. (a) Inversion rate over 30 seeds, all 63 cells; (b) tasks with at least one inverting optimizer. Audited engine, X1/X3 on, K=5, β=2.

the mechanism is not.

The acquisition ranks its hallucinations above designs it holds. The candidate/oracle correction puts the starting designs *inside* the returned pool: every cell is seeded with the top-128 offline designs plus perturbed copies, and gradient and perturbation pool every iterate with that seed set before returning the top 128 by surrogate LCB. A cell returning something worse than it was handed has ranked an invented design above a real one it was holding: an inversion (Figure 3). It is close to ensemble-specific, the ensemble inverting on at least one optimizer on all seven tasks against three for the SVGP and two for the exact GP. The sharpest cell is Branin-2D under ensemble gradient ascent (seven cells in all at unit inversion rate and unit mean fraction worse), where all 30 seeds invert and *every one* of the 128 returned designs scores below the best design the cell started with. This is the elimination-7 localization made concrete, descriptive at n=7, not a mechanism with a causal test. Offline RL names the missing guarantee *safe policy improvement*, that a policy should not be worse than its baseline (Thomas, Theocharous, and Ghavamzadeh 2015; Laroche, Trichelair, and Tachet des Combes 2019; Kumar et al. 2020); offline MBO optimizers ship none, and the inversion is that absence measured.

Three corrections to the prior analysis. We report three in full, including where they fire against us: the released tuning sweep's verdict that the ensemble's gradient collapse is a trust-region-repairable artifact holds on the pre-audit engine but not the audited one, where the collapse is genuine surrogate geometry on most tasks; the collapse is not gradient-specific, so that claim narrows to any aggressive selector; and the premise-coverage 0.97 belongs to a scikit-learn exact GP, not the differentiable GP the grid scores, whose coverage averages 0.831 and falls to 0.38 on Styblinski-5D, both reported with their model (counts, coverages and the $\{\mu - f \leq \beta\sigma\}$ identity in the supplement).

6 Design-Bench Results

The surrogate null. In all four corners the Design-Bench surrogate main effect is essentially zero (η^2_{surr} 0.001–0.032 on five tasks, every estimate at its interval's floor) and the Friedman omnibus fails to reject (p = 0.19 to 0.76): no detectable difference at this power, never equivalence, since at n=7 a paired test needs $|d_z| \geq 1.27$ for 80% power, larger than anything we observe (Agarwal et al. 2021). Nor is it fragile to the task set. Dropping GFP leaves it (0.001–0.021); adding the two MuJoCo tasks makes three of four corners reject but does *not* lift η^2_{surr} off the floor, only to 0.044–0.091, four to nine times below the synthetic 0.405, while the optimizer effect runs 1.6 to 4.5× larger and the one non-rejecting corner carries the *larger* surrogate effect (0.053 against 0.044). Surrogate size does not track rejection; the optimizer axis does, which the added tasks recruit. The null survives the exact-oracle subset (p = 0.34 to 0.51) and a machine-precision noise floor, so neither approximate oracles nor noise manufacture it, though at two such tasks that test is low-powered (supplement). A prior pooling defect (eleven cells, two undefined) is unified to nine before any rank claim (Benavoli, Corani, and Mangili 2016).

Matching the query budget strengthens the optimizer axis. All of that is at native budgets, where gradient outspends perturbation by 6.25× (X3-off) to 11.8× (X3-on), so the optimizer axis could be search intensity in a costume. Equalizing queries across the seven-task grid (Q=25,600 and 51,456; none of 252 cells over 5% off) runs *backwards* to that objection: matching raises the optimizer effect 1.46 to 1.88× in every corner (η^2_{opt} 0.096 → 0.180, 0.145 → 0.255, 0.193 → 0.282, 0.181 → 0.281) while η^2_{surr} *falls* in all four, flipping off_off into rejection so *all four* reject (p = 0.0179 to 0.00057) where three did. The imbalance was *masking* the effect, not manufacturing it. Two limits: it is *established at high budget*, since at the matched low level η^2_{surr} reaches 0.126 above our 0.10 floor and one corner ties (0.085 against 0.081), so the pre-registered primary level decides; and within-corner intervals overlap, licensing point-estimate localization across corners only. The count is the seven-task set, the five-task and GFP-dropped sets rejecting three of four. A native-budget control reproduces the published corners to within 0.004 on every η^2, so the contrast is budget, not call order.

Frozen cells are a benchmark-construction finding. Part of the null is structural, and reusable: several Design-Bench cells are degenerate for surrogate attribution, freezing at the dataset reference so they carry neither a main effect nor an interaction. On TF-Bind-8 the three exact-GP cells return exactly 1.0 on all 16 seeds with zero variance, ensemble-with-perturbation joining them in two corners (four frozen cells of nine), where 1.0 is not a score but the min–max-normalized dataset maximum, already attained by tied rows. On Ant, two of three GP cells return the bit-identical constant 1.52874195557571411 with standard deviation exactly 0.0 across all sixteen seeds, a third (SVGP-with-perturbation) on ten: one value across two surrogate classes is retrieval, not coincidence. An attribution study here must screen for frozen cells first, or read a surrogate axis that cannot move as evidence about surrogates. The mechanism is *LCB paralysis*, not decode snap-back: snap-back needs a decode step, but Ant is continuous with no argmax decode, so our pre-registered kill fires and the GP's lower confidence bound is simply locally maximal at the data. The nearest formalization is an inconsistency result for expected-improvement BO (Yarotsky 2013), differing on three axes, and the trust-region literature uses the same locality *deliberately* (Eriksson et al. 2019), here an unintended terminal state. This does not rescue our positive signal, which stays synthetic-only.

The inversion and the synthetic null are one finding. The section's most awkward number is that the optimizer axis *inverts* here: perturbation leads all four corners (0.76–0.81 against gradient's 0.40–0.68) and η^2_{opt} exceeds η^2_{surr} in point estimate everywhere (intervals overlap, so this is point-estimate localization, never a within-corner separation), while the synthetic optimizer effect is small (η^2_{opt}=0.038 $[0.003, 0.123]$, a magnitude for that grid, not a general verdict). Read through the frozen cells the suites stop disagreeing: if the GP cannot be moved, every optimizer scores it identically, the between-optimizer variance that survives comes from the unfrozen ensemble cells, and the axis inverts as an arithmetic consequence—the payoff of a decomposition over a leaderboard. Four limits: it is two of three Ant GP cells (the gradient cell returns 1.3242 ± 0.1975, unfrozen); fact and label are separable; frozen and near-tied cells are never pooled; and the freeze explains why the axis inverts, not how large the optimizer effect is, which the matched-budget arm establishes.

7 Discussion and Limitations

Pre-registered predictions and their outcomes. Four registered predictions came back against us and two mechanism arms failed; we report all six, since undisclosed they would read as selective. Our registered headline was that the acquisition optimizer explains most of the gap: it does not (η^2_{opt}=0.038 even after correcting the budget imbalance), and the pre-committed contingency, a pivot to the mechanism, is this paper. The ensemble-size reversal criterion did not fire, but its accompanying prediction confirmed: the headline is reported at the K that maximizes it (Confound 3). The β–σ criterion fired, narrowing Confound 4 to its per-task form, and the gradient-specific coverage framing was retracted on our own test. The two mechanism arms cost most: smoothing the mean was built to confirm our published account and refuted it; the phantom-maxima probe was built for a mechanism and returned a seventh elimination.

Interpreting the optimizer effect. Under a matched budget of Q=51,456, $\eta^2_{\text{opt}} = 0.038$ $[0.003, 0.123]$: optimizer choice explains little variance in this grid. It replaces our unmatched 0.005, an eightfold understatement we disclose. The confirmation is not comfortable: the upper bound 0.123 lies inside our pre-registered inconclusive band, so an effect up to $\sim$0.12 is not excluded at n=7, and the secondary level does not corroborate cleanly (0.066 $[0.014, 0.340]$ at Q=4,352). The null is established at high budget, underpowered at low, and the *best* optimizer changes with budget: gradient leads low (0.731), perturbation high (0.732). But the small *marginal* is not a verdict that optimizer choice is arbitrary: its *conditional* effect through the interaction is four to thirty-three times larger, so which optimizer to use depends on the surrogate.

Scope. We make no field reversal: that the statistical model often outranks the acquisition heuristic is a decade-old doctrine (Shahriari et al. 2016); we falsify only the *local* premise of one recent paper—that offline black-box optimization "has focused on improving surrogate models while using fixed search strategies" (Chemingui et al. 2024)—and claim ensemble size as no discovered mechanism (Li, Rudner, and Wilson 2024; Abe et al. 2022), no equivalence on Design-Bench (Agarwal et al. 2021; Lakens 2017), no priority on the K range, and distance-aware uncertainty as not ours (Liu et al. 2020; van Amersfoort et al. 2020). The Design-Bench null is a failure to detect, not equivalence: a two-one-sided-tests bound rejects equivalence at both a 0.5 and a 0.3 margin (supplement) (Lakens 2017). The two protocol arms make the same shape of finding on two axes, correcting them raising the synthetic surrogate effect and matching budgets raising the Design-Bench optimizer effect, both against the reality-check genre's shrink direction (Melis, Dyer, and Blunsom 2018); but that is two instances, not a law, since the ensemble-size confound moves the surrogate effect the other way.

Limitations. Seven synthetic tasks is a small n, binding the corner decomposition, the optimizer interval and the omnibus alike; datasets are drawn once at seed 0, so reported variance is training and optimization, not data-draw. GFP is quarantined from coverage claims for its degenerate decode; excluding it flips one coverage claim (the exact GP's in-distribution coverage from below to above nominal) while the effect-size null is unchanged. The decomposition's direction survives z-score, rank and leave-one-task-out (the headline ranges 0.373–0.485; Griewank-30D, widest in raw range, is the *smallest* lever), and the strengthening survives ϵ^2 bias correction ($0.351 \to 0.395$; supplement). No landscape covariate we tested—dimension, multimodality, separability—predicts the gap at this power (supplement). Two of the seven controls are narrower than the count suggests: capacity is eliminated at fixed depth 2 (the depth form open), and σ at one ensemble construction (a likelihood-trained variant untested). The surviving mechanism is a diagnosis with seven eliminations behind it and no positive causal test in front, both pre-registered positive accounts having returned eliminations, so what stays open is finite: those two arms, plus whether the surrogate-geometry account and the optimizer interaction are one mechanism or two. The positive signal stays synthetic-only.

8 Conclusion

The reported advantage of GP-LCB in offline MBO is measured through five confounds specific to this comparison. Removing the two protocol confounds moves the surrogate main effect from our decomposition of PGS's Table 1 ($\eta^2_{\text{surr}} \approx 0.37$) (Chemingui et al. 2024) to 0.405

$[0.290, 0.556]$: against the direction every named audit in this genre ran, and the net of two corrections against a third, not resolvable at seven tasks. Two firmer results remain: any η^2_{surr} here is a joint artifact of five operating-point coordinates papers do not state, and the surrogate×optimizer interaction, which a one-factor-at-a-time study cannot estimate, exceeds the optimizer main effect in every corner with its interval excluding zero in all four. What the advantage *is* stays open, narrowed by seven controls but not closed. What is measured is the loss—the ensemble's optima worse in true oracle value at matched distance and inflation—and its shape, an acquisition ranking invented designs above real ones it was holding. On the optimizer axis the claim stays narrow: under matched budgets η^2_{opt}=0.038 $[0.003, 0.123]$ *in this grid*, falsifying the local premise of one recent paper (Chemingui et al. 2024) and nothing the standard review settled a decade ago (Shahriari et al. 2016). We offer the taxonomy, the protocol, and the engine stamp that makes either reproducible.

Ethical Statement

This work studies optimization methodology on public benchmarks and introduces no new data or deployed system.

References

Abe, T.; Buchanan, E. K.; Pleiss, G.; Zemel, R.; and Cunningham, J. P. 2022. Deep Ensembles Work, But Are They Necessary? In *Advances in Neural Information Processing Systems (NeurIPS)*.

Agarwal, R.; Schwarzer, M.; Castro, P. S.; Courville, A.; and Bellemare, M. G. 2021. Deep Reinforcement Learning at the Edge of the Statistical Precipice. In *Advances in Neural Information Processing Systems*.

Angelopoulos, A. N.; and Bates, S. 2023. Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning*, 16(4): 494–591.

Balduzzi, D.; Tuyls, K.; Perolat, J.; and Graepel, T. 2018. Reevaluating Evaluation. In *Advances in Neural Information Processing Systems*.

Bellemare, M. G.; Naddaf, Y.; Veness, J.; and Bowling, M. 2013. The Arcade Learning Environment: An Evaluation Platform for General Agents. *Journal of Artificial Intelligence Research*, 47: 253–279.

Benavoli, A.; Corani, G.; and Mangili, F. 2016. Should We Really Use Post-Hoc Tests Based on Mean-Ranks? *Journal of Machine Learning Research*, 17(5): 1–10.

Box, G. E. P.; Hunter, W. G.; and Hunter, J. S. 1978. *Statistics for Experimenters: An Introduction to Design, Data Analysis, and Model Building*. Wiley.

Bressan, P. 2019. Confounds in "Failed" Replications. *Frontiers in Psychology*, 10: 1884.

Brookes, D. H.; Park, H.; and Listgarten, J. 2019. Conditioning by Adaptive Sampling for Robust Design. In *International Conference on Machine Learning*, 773–782. PMLR.

Chemingui, Y.; Deshwal, A.; Hoang, T. N.; and Doppa, J. R. 2024. Offline Model-Based Optimization via Policy-Guided Gradient Search. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Dewolf, N.; De Baets, B.; and Waegeman, W. 2022. Valid Prediction Intervals for Regression Problems. *Artificial Intelligence Review*, 55: 577–613.

Elsayed, K.; and Lacor, C. 2014. Robust Parameter Design Optimization Using Kriging, RBF and RBFNN with Gradient-Based and Evolutionary Optimization Techniques. *Applied Mathematics and Computation*, 236: 325–344.

Eriksson, D.; Pearce, M.; Gardner, J.; Turner, R. D.; and Poloczek, M. 2019. Scalable Global Optimization via Local Bayesian Optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Fannjiang, C.; and Listgarten, J. 2020. Autofocused Oracles for Model-Based Design. In *Advances in Neural Information Processing Systems*, volume 33, 12945–12956.

Ferrari Dacrema, M.; Cremonesi, P.; and Jannach, D. 2019. Are we really making much progress? A worrying analysis of recent neural recommendation approaches. In *Proceedings of the 13th ACM Conference on Recommender Systems (RecSys)*, 101–109.

Gao, L.; Schulman, J.; and Hilton, J. 2023. Scaling Laws for Reward Model Overoptimization. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, 10835–10866. PMLR.

Ghasemipour, K.; Gu, S. S.; and Nachum, O. 2022. Why So Pessimistic? Estimating Uncertainties for Offline RL through Ensembles, and Why Their Independence Matters. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Hansen, N.; Auger, A.; Ros, R.; Mersmann, O.; Tušar, T.; and Brockhoff, D. 2021. COCO: A Platform for Comparing Continuous Optimizers in a Black-Box Setting. *Optimization Methods and Software*, 36(1): 114–144.

Henderson, P.; Islam, R.; Bachman, P.; Pineau, J.; Precup, D.; and Meger, D. 2018. Deep Reinforcement Learning that Matters. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Hutter, F.; Hoos, H.; and Leyton-Brown, K. 2014. An Efficient Approach for Assessing Hyperparameter Importance. In *International Conference on Machine Learning (ICML)*, 754–762.

Jin, Y.; Yang, Z.; and Wang, Z. 2021. Is Pessimism Provably Efficient for Offline Reinforcement Learning? In *International Conference on Machine Learning*, 5084–5096. PMLR.

Jordan, S. M.; White, A.; Castro, P. S.; Bellemare, M. G.; Machado, M. C.; and Precup, D. 2024. Position: Benchmarking is Limited in Reinforcement Learning Research. In *International Conference on Machine Learning*.

Kim, M.; Gu, J.; Yuan, Y.; Yun, T.; Liu, Z.; Bengio, Y.; and Chen, C. 2026. Offline Model-Based Optimization: Comprehensive Review. *Transactions on Machine Learning Research*. Survey Certification.

Krishnamoorthy, S.; Mashkaria, S. M.; and Grover, A. 2023. Diffusion Models for Black-Box Optimization. In *International Conference on Machine Learning*, 17842–17857. PMLR.

Kumar, A.; Zhou, A.; Tucker, G.; and Levine, S. 2020. Conservative Q-Learning for Offline Reinforcement Learning. In *Advances in Neural Information Processing Systems*, volume 33, 1179–1191.

Lakens, D. 2017. Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, 8(4): 355–362.

Lakshminarayanan, B.; Pritzel, A.; and Blundell, C. 2017. Simple and Scalable Predictive Uncertainty Estimation using Deep Ensembles. In *Advances in Neural Information Processing Systems*, volume 30.

Laroche, R.; Trichelair, P.; and Tachet des Combes, R. 2019. Safe Policy Improvement with Baseline Bootstrapping. In *International Conference on Machine Learning*.

Li, Y. L.; Rudner, T. G. J.; and Wilson, A. G. 2024. A Study of Bayesian Neural Network Surrogates for Bayesian Optimization. In *International Conference on Learning Representations*.

Liang, Q.; Gongora, A. E.; Ren, Z.; Tiihonen, A.; et al. 2021. Benchmarking the Performance of Bayesian Optimization across Multiple Experimental Materials Science Domains. *npj Computational Materials*, 7: 188.

Liu, J. Z.; Lin, Z.; Padhy, S.; Tran, D.; Bedrax-Weiss, T.; and Lakshminarayanan, B. 2020. Simple and Principled Uncertainty Estimation with Deterministic Deep Learning via Distance Awareness. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Lu, C.; Ball, P. J.; Parker-Holder, J.; and Roberts, S. 2022. Revisiting Design Choices in Offline Model-Based Reinforcement Learning. In *International Conference on Learning Representations*.

Lucic, M.; Kurach, K.; Michalski, M.; Gelly, S.; and Bousquet, O. 2018. Are GANs Created Equal? A Large-Scale Study. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Maassen, E.; van Assen, M. A. L. M.; Nuijten, M. B.; Olsson-Collentine, A.; and Wicherts, J. M. 2020. Reproducibility of Individual Effect Sizes in Meta-Analyses in Psychology. *PLOS ONE*, 15(5): e0233107.

Melis, G.; Dyer, C.; and Blunsom, P. 2018. On the State of the Art of Evaluation in Neural Language Models. In *International Conference on Learning Representations (ICLR)*.

Moosbauer, J.; Binder, M.; Schneider, L.; Pfisterer, F.; Becker, M.; Lang, M.; Kotthoff, L.; and Bischl, B. 2022. Automated Benchmark-Driven Design and Explanation of Hyperparameter Optimizers. *IEEE Transactions on Evolutionary Computation*, 26(6): 1336–1350.

Musgrave, K.; Belongie, S.; and Lim, S.-N. 2020. A Metric Learning Reality Check. In *European Conference on Computer Vision (ECCV)*, 681–699.

NIST/SEMATECH. 2012. NIST/SEMATECH e-Handbook of Statistical Methods. https://www.itl.nist.gov/div898/handbook/. Section 5: Process Improvement.

Recht, B.; Roelofs, R.; Schmidt, L.; and Shankar, V. 2019. Do ImageNet Classifiers Generalize to ImageNet? In *International Conference on Machine Learning (ICML)*, 5389–5400.

Shahriari, B.; Swersky, K.; Wang, Z.; Adams, R. P.; and de Freitas, N. 2016. Taking the Human Out of the Loop: A Review of Bayesian Optimization. *Proceedings of the IEEE*, 104(1): 148–175.

Srinivas, N.; Krause, A.; Kakade, S. M.; and Seeger, M. W. 2012. Gaussian Process Optimization in the Bandit Setting: No Regret and Experimental Design. *IEEE Transactions on Information Theory*, 58(5): 3250–3265.

Stanton, S.; Maddox, W.; and Wilson, A. G. 2023. Bayesian Optimization with Conformal Prediction Sets. In *International Conference on Artificial Intelligence and Statistics*. PMLR.

Tan, R.-X.; Xue, K.; Lyu, S.-H.; Shang, H.; Wang, Y.; Wang, Y.; Fu, S.; and Qian, C. 2025. Offline Model-Based Optimization by Learning to Rank. In *International Conference on Learning Representations*.

Thomas, P. S.; Theocharous, G.; and Ghavamzadeh, M. 2015. High Confidence Policy Improvement. In *International Conference on Machine Learning*.

Trabucco, B.; Geng, X.; Kumar, A.; and Levine, S. 2022. Design-Bench: Benchmarks for Data-Driven Offline Model-Based Optimization. In *International Conference on Machine Learning*, 21658–21676. PMLR.

Trabucco, B.; Kumar, A.; Geng, X.; and Levine, S. 2021. Conservative Objective Models for Effective Offline Model-Based Optimization. In *International Conference on Machine Learning*, 10358–10368. PMLR.

van Amersfoort, J.; Smith, L.; Teh, Y. W.; and Gal, Y. 2020. Uncertainty Estimation Using a Single Deep Deterministic Neural Network. In *International Conference on Machine Learning (ICML)*, 9690–9700.

Yarotsky, D. 2013. Examples of Inconsistency in Optimization by Expected Improvement. *Journal of Global Optimization*, 56(4): 1773–1790.

Zhu, Y.; Qian, H.; et al. 2025. SOO-Bench: Benchmarks for Evaluating the Stability of Offline Black-Box Optimization. In *International Conference on Learning Representations*.